Quadratic Studio

Moldash Bekarys · Quadratic equations · Individual project

One concept. Three starting points.

A responsive learning lab develops one core task: finding and interpreting the zeros of a quadratic. A shared seed connects three differentiated versions; eight seeds provide 24 tasks. Learners choose their level, receive progressive guidance and check their understanding.

LEVEL	EXAMPLE & VERIFIED RESULT	DIFFERENTIATION
Foundation	$x^2 - 5x + 6 = 0$ Roots: 2 and 3	Integer factors; visible three-step scaffold; direct hints and zero-product reasoning.
Standard	$2x^2 - 10x + 9 = 0$ Roots: $(5 \pm \sqrt{7}) / 2$	Non-monic equation; irrational roots; discriminant and quadratic formula; less scaffolding.
Advanced	$h(t) = -5t^2 + 25t + 30$ Landing: 6 s; maximum: 61.25 m	Physical domain and units; reject negative time; evaluate the vertex at $t = 2.5$ s.

AI architecture

Transformers.js runs the open-source SmolLM2-135M-Instruct model (q4) in a browser Web Worker using ONNX WASM. The first coach request downloads model files; inference runs on-device. No API key or external inference backend is required; questions are not uploaded. Deterministic code transforms parameters and checks answers. The model adapts explanations using verified mathematical context.

Prompt design

Abridged system template: “You are a patient mathematics coach in Quadratic Studio. [Level-specific style.] Use at most 70 words. Adapt the supplied VERIFIED GUIDANCE to the learner question. Never invent coefficients, computations, answers or extra tasks. [Give a hint rather than the final answer when the walkthrough is closed.]”

Foundation requests simple words and one small next step. **Standard** requests curriculum vocabulary and a useful question about coefficients, Δ or ± cases. **Advanced** requests units, the physical domain and multi-step reasoning. Context includes the current equation, level, learner question, verified hint and last checking feedback. Final results are supplied only with an open walkthrough. Exact prompts: **src/math.js**.

Feedback, validation and limitations

Immediate checking accepts fractions, radicals and decimals within 0.01, and explains notation errors, duplicate roots and inadmissible negative time. Hints, complete solutions and an interactive graph support self-paced learning. Progress is stored locally. Eight test groups verify all 24 tasks, parsing and prompt context; browser checks exercise learning interactions and the honestly labelled model-unavailable fallback. Real inference and public publication remain unverified where network access is blocked. AI wording may be imperfect; the verified walkthrough remains authoritative.

Development disclosure: Prepared with AI assistance; mathematics verified independently of model output.
References: huggingface.co/docs/transformers.js · huggingface.co/onnx-community/SmolLM2-135M-Instruct · vite.dev/guide/static-deploy.html · GitHub Pages documentation.